

Question

A compound **A** with the molecular formula C_8H_{14} contains a quaternary carbon. After heating and refluxing it with $KMnO_4$, only a product with the molecular formula $C_7H_{12}O$ is detected in the reaction solution. **A** reacts with hydrobromic acid to give **B**, while reaction with hydrobromic acid in the presence of peroxide gives isomer **C**. **B** in sodium ethoxide-ethanol gives **D**. **D** reacts with BH_3 and is further treated with acetic acid to give compound **E** containing a five-membered ring structure. **D** can decolorize Br_2 and produce **F**. Treatment of **F** with sodium ethoxide-ethanol solution yields two isomeric products **G** and **H**. Deduce the structures of all species based on the above information.

Based on your deduced results, choose the correct option. If all options are incorrect, select option A: All other options are incorrect.

- A.** All other options are incorrect.
- B.** Compound **A** has 5 different chemical shift hydrogens in the nuclear magnetic resonance hydrogen spectrum.
- C.** Among compounds **A** ~ **H**, there are 4 types of hydrocarbons.
- D.** There are 2 pairs of isomers among compounds **A** ~ **H**
- E.** The substituents on the ring in compound **G/H** are completely identical to those in compound **A**.
- F.** The different kinds of substituents on the ring in compound **A** are located on adjacent C atoms.

Answer

Correct Answer: **A**

Detailed Explanation

To solve this problem, we first need to deduce the structure of **A** based on the known information and the reaction properties of **A**.

The molecular formula of **A** is C_8H_{14} , and its degree of unsaturation $\Omega = 2$.

CHECKPOINT

0.5 PTS

The degree of unsaturation of **A** is $\Omega = 2$.

This indicates that it contains an unsaturated $C = C$ double bond or a cyclic structure. Since **A** reacts with $KMnO_4$ under reflux with heating, only a product with the molecular formula $C_7H_{12}O$ is detected in the reaction solution. The ratio of the species after the reaction shows that **A** has one less carbon, which indicates that it contains only one unsaturated $C = C$ double bond and it is a terminal alkene double bond (``[[CX4][CX3](=[CH2])[CX4][`]`, hereinafter referred to as ``[CC(=C)C[`]` for short), and the other degree of unsaturation is contributed by a cyclic structure.

CHECKPOINT

0.5 PTS

A contains a terminal alkene double bond and a cyclic structure

A reacts with hydrobromic acid to give **B**, then the product after the latter tertiary carbocation combines with a bromide ion is ``[CC(Br)(C)C[`]`.

CHECKPOINT

0.5 PTS

B contains the fragment []CC(Br)(C)C[]

The reaction with hydrobromic acid in the presence of peroxide gives the isomer **C**. This is a free radical reaction, and the direction of hydrobromic acid addition to the double bond is opposite to that of the ionic reaction. The product obtained should be []CC(CBr)C[].

CHECKPOINT

0.5 PTS

C contains the fragment []CC(CBr)C[]

B gives **D** in sodium ethoxide-ethanol. This is a one-step elimination reaction. It should be noted that **A** cannot be generated here, but an isomer **D** with a different double bond position from **A** should be generated. Therefore, the structure of **D** can be written as []C=C(C)C[].

CHECKPOINT

0.5 PTS

D contains the fragment []C=C(C)C[]

D reacts with BH_3 and is further treated with acetic acid to obtain a compound **E** containing a five-membered ring structure. This step is a hydroboration reduction reaction, which reduces the unsaturated double bond to a saturated bond. Obviously, no ring formation has occurred. Therefore, the "five-membered ring" described here was originally present in the substrate. This also confirms the assumption that "**A** has a cyclic structure". Therefore, the cyclic structure here is a five-membered ring. The structure of **E** contains a cyclic structure like C1C(C)CCC1, while **A** has an additional terminal alkene double bond outside the ring C1C(=C)CCC1.

CHECKPOINT

1 PTS

E contains the `C1C(C)CCC1` structure, and **A** contains the `C1C(=C)CCC1` structure

D can decolorize Br_2 and generate **F**. This step is a dibromination of an alkene. According to the previous inference, **F** contains the `C1(Br)C(Br)(C)CCC1` structure.

CHECKPOINT

0.5 PTS

F contains the `C1(Br)C(Br)(C)CCC1` structure

F is treated with sodium ethoxide-ethanol solution to generate two isomeric products **G** and **H**. This is a one-step elimination reaction to obtain a diene. Rewriting `C1(Br)C(Br)(C)CCC1` as the equivalent `[C:1]1[C:2]([Br:a])[C:3]([Br:b])[C:4][C:5][C:6]1` indicates that the hydrogen on `[C:1]` can undergo elimination with `[Br:a]` and there is only one way, which indicates that `[C:1]` is not the quaternary carbon in **A** as indicated in the question. Similarly, the hydrogen on `[C:4]/[C:5]` can undergo two different elimination modes with `[Br:b]` to obtain two isomeric dienes, which also indicates that `[C:4]/[C:5]` is not a quaternary carbon either. Therefore, only `[C:6]` can be a quaternary carbon. Combining with the molecular formula C_8H_{14} of **A** in the question, it is not difficult to find that the quaternary carbon is connected to two methyl groups. Therefore, the structure of **A** is `C1C(=C)CC(C)(C)C1`.

CHECKPOINT

3 PTS

The structure of **A** is `C1C(=C)CC(C)(C)C1`

Based on this, we can get:

- The structure of **B** is `C1C(CBr)CC(C)(C)C1`
- The structure of **C** is `C1C(Br)(C)CC(C)(C)C1`

CHECKPOINT

1 PTS

The structure of **B** is `C1C(CBr)CC(C)(C)C1`, and the structure of **C** is `C1C(Br)(C)CC(C)(C)C1`

- There are two possibilities for the structure of **D**: `C1=C(C)CC(C)(C)C1` or `C1C(C)=CC(C)(C)C1`. If it is the latter, then its generated **F** cannot undergo an elimination reaction under the condition of sodium ethoxide to obtain two different elimination products (only one bromine can be eliminated, and there are three possible products), so it should be `C1=C(C)CC(C)(C)C1`

CHECKPOINT

3 PTS

The structure of **D** is `C1=C(C)CC(C)(C)C1`

- The structure of **E** is `C1C(C)CC(C)(C)C1`
- The structure of **F** is `C1(Br)C(Br)(C)CC(C)(C)C1`
- The structure of **G/H** is `C1=CC(C)=CC1(C)C`/`C1=CC(=C)CC1(C)C`

Finally, judge each option one by one:

- Compound **A** has 6 different chemical shift hydrogens in the nuclear magnetic resonance hydrogen spectrum (2 types of terminal alkene + 3 types of -CH₂- in the ring + 1 type of gem-dimethyl), so option B is incorrect.

CHECKPOINT

1 PTS

Compound **A** has 6 different chemical shift hydrogens in the nuclear magnetic resonance hydrogen spectrum

- There are 5 types of hydrocarbons among compounds **A** ~ **H** (**A**/**D**/**E**/**G**/**H**), so option C is incorrect.

CHECKPOINT

1 PTS

There are 5 types of hydrocarbons among compounds **A** ~ **H**

- There are 3 pairs of isomers among compounds **A** ~ **H** (**A** – **D**/**B** – **C**/**G** – **H**), so option D is incorrect.

CHECKPOINT

1 PTS

There are 3 pairs of isomers among compounds **A** ~ **H**

- Compound **A** has 2 methyl groups and 1 methylene group on the ring, and compound **G**/**H** has only one that is the same as **A**, and all the substituents on the other ring are methyl groups. Option E is incorrect.

CHECKPOINT

1 PTS

The substituents on the ring in compound **G**/**H** are not all the same as those on the ring in compound **A**

- The 2 methyl groups and the methylene group on the ring in compound **A** are located on spaced C atoms.
Option F is incorrect.

CHECKPOINT

1 PTS

The 2 methyl groups and the methylene group on the ring in compound **A** are located on spaced C atoms

All options are incorrect, choose option A.